

RWA, Impairment, and Risk-Sensitive Pricing

Cheat sheet — running example: unsecured consumer loan €10,000, PD = 2%, LGD = 45%, 3-year annuity at 8.5%.

1. Foundations

Standard normal: $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, $\Phi(x) = \int_{-\infty}^x \varphi(z) dz$, $G = \Phi^{-1}$.

Memorise: $G(0.02) = -2.0537$, $G(0.999) = 3.0902$.

Lemma (linear combinations). $Y, \varepsilon \sim \mathcal{N}(0, 1)$ indep. $\Rightarrow aY + b\varepsilon \sim \mathcal{N}(0, a^2 + b^2)$ (proof: characteristic functions).

Conditional. $\mathbb{P}(A | B) = \mathbb{P}(A \cap B) / \mathbb{P}(B)$; $X = aY + b\varepsilon | Y = y \sim \mathcal{N}(ay, b^2)$.

Tail vs. expectation: capital $\approx$ tail quantile of L minus $EL = \mathbb{E}[L]$. EL is priced/provisioned; unexpected loss is capitalised.

Annuity: PV = N gives $A = \frac{Nr}{1 - (1+r)^{-n}}$; $N = 10,000$, $r = 8.5\%$, $n = 3$: $A = e3,915.39$; balances $10,000 \rightarrow 6,934.61 \rightarrow 3,608.66$.

2. Part A — RWA under A-IRB (CRR)

Chain: PD, LGD, EAD $\rightarrow K \rightarrow RWA = 12.5 K EAD \rightarrow$ own funds = $8\% \times RWA$.

Thm (scaling). own funds = $KEAD = 0.08 RWA \Rightarrow RWA = KEAD/0.08$; the 12.5 is pure algebra.

ASRF factor model:

$$X_i = \sqrt{R}Y + \sqrt{1-R}\varepsilon_i, \quad \text{Var}(X_i) = 1, \quad \text{Cov}(X_i, X_j) = R.$$

Default threshold (Merton): $PD = \mathbb{P}(X_i < c) = \Phi(c)$, so $c = G(PD)$.

Conditional PD:

$$\mathbb{P}(\text{def} | Y = y) = \Phi\left(\frac{G(PD) - \sqrt{R}y}{\sqrt{1-R}}\right).$$

Stress PD (adverse state $y = G(0.999)$):

$$PD^{\text{stress}} = \Phi\left[\frac{G(PD)}{\sqrt{1-R}} + \sqrt{\frac{R}{1-R}} G(0.999)\right]$$

Gordy formula (capital factor):

$$K = \text{LGD}(PD^{\text{stress}} - PD) = \text{LGD}\left[\Phi(\dots) - PD\right]$$

Capital = $\text{LGD} \times$ excess of stress PD over PD ($EL = \text{PDLGD}$ subtracted; stress loss covers the 99.9% tail).

Gordy's thm (sketch). Conditional on Y , defaults are independent; $n \rightarrow \infty$ makes the default rate converge to $PD^{\text{stress}}(Y)$ a.s. $\Rightarrow$ capital depends only on (PD, LGD, R) , not portfolio composition (portfolio invariance).

CRR retail correlation (Art. 154):

$$A(PD) = \frac{1 - e^{-35PD}}{1 - e^{-35}}, \quad R(PD) = 0.16 - 0.13 A(PD)$$

anchors $R \rightarrow 16\%$ ($PD \rightarrow 0$), $R \rightarrow 3\%$ ($PD \rightarrow 1$); constants $R = 0.15$ (mortgages), $R = 0.04$ (revolving retail).

Running example: $A = 0.5034$, $R = 9.46\%$; $z = -2.1561 + 1.0000 = -1.1597$; $PD^{\text{stress}} = 12.31\%$; $K = 0.45 \times 0.1031 = 4.64\%$; $RW = 58.0\%$; $RWA = e5,798.64$; own funds = $e463.89$; $EL = e90.00$.

Sensitivities: K concave in PD (falling R cushions); RWA linear in EAD , LGD . Example: $PD = 8\% \Rightarrow R = 3.79\%$,

$PD^{\text{stress}} = 20.64\%$, $K = 5.69\%$, $RW = 71.1\%$, own funds $e568.64$, $EL = e360$.

3. Part B — Impairment under IFRS 9

Stages:

1	performing	12m ECL; interest gross
2	SICR	lifetime ECL; interest gross
3	credit-impaired	lifetime ECL; interest net

ECL formula (marginal PDs, discount at original EIR):

$$ECL = \sum_{t=1}^T m_t \text{LGD} E_t (1 + \text{EIR})^{-t}$$

m_t = conditional-on-survival one-year PD; equivalently $(S_t - S_{t-1})$ with $S_t = 1 - \prod_{j \leq t} (1 - m_j)$.

Thm (monotonicity). lifetime $ECL \geq 12m$ ECL, strict if any $m_t > 0$, $t \geq 2 \Rightarrow$ stage-2 migration is a P&L cliff (allowance never decreases on migration).

Running example: $m = (2\%, 2.5\%, 3\%)$, $E = (10,000; 6,934.61; 3,608.66)$:

12m: $\frac{0.02 \cdot 0.45 \cdot 10,000}{1.085} = e82.95$; lifetime: $82.95 + 66.27 + 38.14 = e187.36$

Scenario weighting (weights 20/60/20, PDs halved/base/doubled; ECL linear in m):

$$ECL_{\text{lt}} = 0.2(93.68) + 0.6(187.36) + 0.2(374.72) = e206.10$$

weighted 12m: $e91.24$.

EIR subtlety: higher EIR $\Rightarrow$ lower discounted allowance (12m ECL: $85.71/82.95/80.36/75.00$ at $5/8.5/12/20\%$).

Reconciliation: regulatory EL = undiscounted year-1 ECL term ($e90$); capital covers the tail ($e463.89$), IFRS 9 books the expectation ($e82.95 - e206.10$).

4. Part C — Risk-Sensitive Pricing (RAROC)

$$\text{RAROC} = \frac{\text{revenue} - \text{funding cost} - \text{opex} - \text{EL}}{\text{capital}}$$

Pricing rule: $\text{RAROC} \geq \text{COE}$; break-even rate r_L where $\text{RAROC} = \text{COE} = h$. Solving:

$$r_L = r_F + \frac{O + \text{EL}_{\text{p.a.}} + hC}{B} \quad C = 0.08 RWA$$

f = funding + opex + EL + cost of capital (Parts B and A in price).

Running example: $r_F = 2\%$, $O = e166.67$, $EL = e90$, $C = e463.89$, $h = 12\%$, $B = e6,847.75$:

$$r_L = 0.02 + \frac{166.67 + 90 + 55.67}{6,847.75} = 6.56\%$$

margin 4.56%: opex 2.43% + EL 1.31% + capital 0.81%.

Cross-check: $\text{RAROC}(8.5\%) = \frac{188.44}{463.89} = 40.6\%$ — inflated because regulatory capital for low-PD retail is small; professional fixes: economic capital (3–5 $\times$), Pillar 2, acquisition costs.

Riskier borrower (PD = 8%): $r_L = 0.02 + \frac{166.67+360+68.24}{6,847.75} = 10.69\%$ — EL does most of the pricing work.
Lifetime-EL basis: $EL = 206.10/3 = e68.70$ p.a. $\Rightarrow r_L \approx 6.25\%$ (12m basis is the conservative one).

5. Synthesis: one loan, three numbers

Tail capital	A-IRB	€463.89 (RWA €5,798.64)
Provision	IFRS 9	€82.95–€206.10
Break-even	RAROC	6.56% p.a.

Same (PD, LGD, EAD) drive all three; capital rises modestly with risk (correlation cushion), allowance and price rise roughly with EL.

6. Dashboards

RWA_visualised.py (5001): *RW*, *K*, *R* vs PD; RWA vs EAD.
 Impairment_visualised.py (5002): stage cliff, scenario weights, per-year ECL.
 Pricing_visualised.py (5003): break-even decomposition, RAROC vs hurdle, capital multiplier, EL basis.

Key tables (compressed)

Break-even rate (%), hurdle *h* × capital multiplier:

Mult.	12m EL			lifetime EL		
	8%	12%	16%	8%	12%	16%
1.0	6.29	6.56	6.83	5.98	6.25	6.52
2.0	6.83	7.37	7.92	6.52	7.06	7.60
4.0	7.92	9.00	10.08	7.60	8.69	9.77

PD sensitivity (LGD 45%, EAD €10,000):

PD	<i>R</i>	PD ^{stress}	<i>K</i>	<i>RW</i>	RWA	req. rate
0.5%	13.9%	6.25%	2.59%	32.4%	3,236	5.22%
1%	12.2%	9.14%	3.66%	45.8%	4,577	5.73%
2%	9.5%	12.31%	4.64%	58.0%	5,799	6.56%
5%	5.3%	16.81%	5.31%	66.4%	6,642	8.65%
8%	3.8%	20.64%	5.69%	71.1%	7,108	10.69%
15%	3.1%	30.75%	7.09%	88.6%	8,860	15.53%
30%	3.0%	50.44%	9.20%	115%	11,498	25.76%

Educational summary with didactic simplifications — not regulatory, accounting, or financial advice. Full derivations, proofs, and sources: *Understanding RWA, Impairment, and Risk-Sensitive Pricing*.